HR Attrition Analytics — Executive Report

Insights, Risk Concentration, Drivers & Retention Recommendations

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Tools: Python, SQL, Power BI, DAX
Dataset: 1,470 employees

Executive Overview

The workforce contains **1,470 employees**, of whom **237 have left**, producing an overall attrition rate of **16.12%**. The analysis answers three management questions: who is leaving, why they are leaving, and where organizational risk is concentrated.

Key executive findings

• Early-career employees (18–25) show 35.77% attrition.
• Single employees show 25.53% attrition.
• Overtime is associated with 30.53% attrition versus 10.44% without overtime.
• Job involvement declines from 33.73% attrition at Low to 9.03% at Very High.
• Sales has the highest displayed departmental rate at 20.63%.
• Sales Representative is the highest-risk displayed role at 39.76%.
• Entry Level has the highest displayed job-level rate at 26.34%.
• Laboratory Technician contributes 62 leavers, or 26.16% of all leavers.
• The Up to $5K salary band contributes 68.78% of all leavers.

All findings are observational associations. They should be validated by HR before being interpreted as causal relationships.

1. Who Is Leaving?

Age profile

Age group	Attrition rate
18–25	35.77%
26–35	19.14%
36–45	9.19%
46–55	11.50%
55+	17.02%

The age pattern is not a simple linear decline: attrition peaks at 18–25, bottoms at 36–45, then rises again at 55+. These should be treated as potentially different retention problems.

Marital status: Single employees show 25.53% attrition versus 12.48% for married and 10.09% for divorced employees.

Gender: the dashboard displays workforce composition (588 female, 882 male), not gender-specific attrition rates. A gender attrition-rate measure is required before drawing a driver conclusion.

Education field: Human Resources (25.93%), Technical Degree (24.24%), and Marketing (22.01%) have the highest displayed rates.

2. Why Are They Leaving?

Overtime

Overtime	Attrition rate
Yes	30.53%
No	10.44%

The gap is substantial and makes overtime an important workload/staffing signal for further investigation.

Job involvement

Level	Attrition rate
Low	33.73%
Medium	18.93%
High	14.40%
Very High	9.03%

This is one of the clearest gradients in the dashboard: attrition decreases consistently as job involvement increases.

Distance from home: the displayed bands range from 13.55% at the lowest band to 22.14% at the highest displayed band, suggesting a positive relationship between commute distance and attrition.

Work-life balance, environment satisfaction, and job satisfaction also show higher displayed attrition at weaker employee-experience levels.

3. Compensation Segmentation

Salary band	Employees	Leavers	Rate	Share of leavers
Up to $5K	749	163	21.76%	68.78%
$5K–$10K	440	49	11.14%	20.68%
$10K–$15K	148	20	13.51%	8.44%
$15K+	133	5	3.76%	2.11%

The Up to $5K band is the dominant exposure group. The $10K–$15K band shows a different pattern: leavers have longer tenure and more years since promotion, suggesting a career-progression hypothesis rather than a single compensation story.

4. Where Is the Organizational Risk?

Department	Attrition rate
Sales	20.63%
Human Resources	19.05%
Research & Development	13.84%

Sales has the highest displayed departmental rate.

Job role

Job role	Attrition rate
Sales Representative	39.76%
Laboratory Technician	23.94%

Job role	Attrition rate
Human Resources	23.08%
Sales Executive	17.48%
Research Scientist	16.10%

Job level

Job level	Attrition rate
Entry Level	26.34%
Senior Level	14.68%
Mid Level	9.74%
Director Level	7.25%
Manager Level	4.72%

Business travel

Travel pattern	Attrition rate
Travel Frequently	24.91%
Travel Rarely	14.96%
Non-Travel	8.00%

5. Rate vs. Exposure — The Critical Management Distinction

Sales Representative has the highest displayed rate (39.76%), but Laboratory Technician contributes 62 actual leavers, or 26.16% of all leavers. R&D; has a lower departmental rate (13.84%) but accounts for 56.12% of all leavers because of its workforce size.

Retention resources should therefore be allocated using both risk intensity and organizational exposure.

6. Cross-Segment Risk

Risk combination	Observed attrition	Sample
Sales Representative + Overtime	66.7%	n=24
Sales Representative + Frequent Travel	65.2%	n=23

These are high-priority risk signals, but their sample sizes are small. They should trigger targeted validation rather than be treated as proof of causality.

7. Strategic Recommendations

1. **Sales workload review:** investigate overtime, staffing, workload and travel within Sales Representative roles.
2. **Department-specific entry-level retention:** strengthen onboarding, mentoring and progression in high-risk departments rather than applying one generic program.
3. **Prioritize by rate and exposure:** combine attrition rate, number of leavers, share of leavers and workforce size.
4. **Lower-salary retention review:** evaluate compensation, tenure, job level, promotion and overtime for the Up to $5K segment.
5. **Career-path review:** identify long-tenured employees with extended promotion gaps, especially in the $10K–$15K band.
6. **Travel and work-life review:** investigate frequent travel and poor work-life balance alongside overtime and role

design.

8. Management Monitoring KPIs

Overall attrition • Department attrition • Role attrition • Job-level attrition • New-hire attrition • Overtime attrition • Travel-related attrition • Salary-band attrition • High-risk segment exposure • Leavers by segment • Share of total leavers • Tenure of leavers • Years since last promotion.

9. Analytical Limitations

The dataset is observational and does not establish why employees left. Some high-risk combinations have small samples. Attrition rate must be interpreted together with population size. The dashboard does not capture all possible causal factors such as management quality, external labor-market conditions or personal circumstances.

10. Final Executive Conclusion

The 16.12% company-wide attrition rate hides several distinct retention problems. The evidence points toward targeted intervention: early-career retention, workload/overtime, Sales role risk, entry-level risk, lower-salary exposure, career progression and travel/work-life burden.

Recommended management principle: intervene where elevated attrition, meaningful workforce exposure and actionable operational factors overlap, then monitor whether the intervention changes the outcome.